

Problem Set 3: Producer Theory, Costs & Technology

Microeconomic Theory for Experimentalists & Applied Microeconomists

Due: [date — before Lecture 1 of Module 4]

Ground rules: collaboration on ideas is encouraged; write-ups are individual. You may use AI tools for parts flagged with ⟨AI⟩ and must attach the transcript. Show all calculations.

Weights: Part A — 30 points. Part B — 45 points. Part C — 25 points.

Part A: Theory — Guided Calculations (30 points)

Question 1 (Cost minimization and cost curves — 15 points). *A firm's technology is $q = \sqrt{KL}$. The wage is $w = 4$ and the capital rental rate is $r = 1$. Inputs are continuously divisible.*

- (a) (4 pts) *At the cost-minimizing input mix, the marginal rate of technical substitution $MP_L/MP_K = K/L$ equals the input price ratio w/r . Use this and the production target $q = 10$ to show the firm hires $L = 5$ and $K = 20$, at cost 40.*
- (b) (4 pts) *Repeat the logic for a general output target q to show the long-run cost function is $C(q) = 4q$. What are long-run average and marginal cost? In one sentence, connect the shape of $C(q)$ to this technology's returns to scale.*
- (c) (5 pts) *Short run: capital is fixed at $\bar{K} = 20$. Show that short-run cost is $SRC(q) = q^2/5 + 20$, that short-run average cost $SRAC = q/5 + 20/q$ is minimized at exactly $q = 10$ with $SRAC = 4$, and that short-run marginal cost equals $SRAC$ at that point. Hint: to minimize $SRAC$, set its derivative to zero — or use the fact that MC cuts AC at AC's minimum. State in one sentence what this illustrates about the relation between short-run and long-run cost curves.*
- (d) (2 pts) *An artefactual experiment hands manager-subjects this exact technology and these prices and asks them to produce $q = 10$ (all subjects reach the output target). Subjects' average spending is 46, falling to 41 by the tenth round. In two sentences: what economic quantity does the gap from 40 measure, and why does its decline with feedback matter for interpreting field productivity dispersion?*

Question 2 (Technology adoption as an NPV decision — 15 points). *Fertilizer costs \$10 per acre, paid at planting. It raises harvest revenue by \$15 per acre, received one season after planting. A farmer is deciding whether to adopt on one acre.*

- (a) (4 pts) *Compute the NPV of adoption for a farmer whose seasonal discount rate is 10%. Then compute the break-even seasonal discount rate — the rate at which NPV is exactly zero — and say in one sentence why this number makes low adoption a puzzle.*

- (b) **(4 pts)** *Credit: the farmer has no cash at planting and must borrow the \$10 at seasonal interest rate b , repaying at harvest. For $b = 30\%$: does adoption still pay? Find the break-even borrowing rate. What does the comparison of the two break-even rates in (a) and (b) tell you about how much of the adoption puzzle credit constraints alone can explain?*
- (c) **(4 pts)** *Risk: harvest gains are actually +\$25 in a good year and +\$5 in a bad year, each with probability $1/2$ (same mean, \$15). A risk-neutral farmer's decision is unchanged — say why in one sentence. Now consider a “safety-first” farmer who refuses any option with a possible net loss: what does she do, and what intervention does this friction respond to that credit and information frictions do not?*
- (d) **(3 pts)** *A survey finds non-adopting farmers who can state the fertilizer return correctly and who hold liquid savings exceeding \$10 at planting. Which frictions does this observation rule out, which remain, and what is the cheapest experiment that would test the leading remaining candidate? (Hint: think about WHEN the purchase decision is made.)*

Part B: Experimental Design (45 points)

Question 3 (Separating the adoption frictions). *Design a field experiment in a smallholder-agriculture setting whose goal is to separate at least three of the four adoption frictions — credit constraints, information/learning, risk, and present bias/procrastination — for a positive-NPV input (fertilizer or equivalent). A price-subsidy arm alone is not acceptable: as discussed in class, a price cut raises adoption under every friction. Specify:*

- (a) **(6 pts)** *Setting, subject pool, and recruitment, and why the input's return is credibly positive for this population (your design should say what evidence anchors the NPV premise).*
- (b) **(12 pts)** *Treatment arms. For each arm, state which friction it targets and — the key requirement — which pattern of effects across arms identifies each friction. Include at least one timing manipulation in the spirit of Duflo, Kremer & Robinson (2011). Say explicitly what the DKR-style comparison (small early discount vs. large later subsidy) identifies that neither arm alone does.*
- (c) **(9 pts)** *Outcomes and measurement: what you measure, from what source (administrative purchase records vs. self-reports), at what times, and one measurement threat with your answer to it.*
- (d) **(9 pts)** *Hypotheses and rough sample-size reasoning: state the adoption rates you expect by arm under each friction story (anchor the control-arm rate and the plausible effect sizes on the DKR setting), and the approximate n per arm. Simulation-based reasoning is welcome; formulas are not required.*
- (e) **(9 pts)** *The biggest confound in YOUR design and your answer to it. (Candidates: seasonal liquidity correlates with nearly everything; treatment arms that change the number of contacts with enumerators; spillovers between neighbors assigned to different arms.)*

Part C: Silicon Pilot Interpretation <AI> (25 points)

Question 4 (The managers that always adopt). *You run `module3-simulation-lab.ipynb`: LLM plant managers decide on the \$10-at-planting / \$15-at-harvest input under three personas (patient*

planner, impatient procrastinator, cautious skeptic) and two decision timings (cash-in-hand vs. cash-tight). Adoption comes back at **roughly 96% in every cell**: the timing manipulation moves it by one percentage point, and the “impatient procrastinator” adopts as readily as the “patient planner.” In the human literature, baseline adoption is a minority and DKR’s timing manipulation moved it substantially.

- (a) **(8 pts)** Give two mechanisms that could produce this pattern, at least one specific to LLMs. (Think about what a transparently positive NPV calculation does to a model trained to be helpful and numerate, and about whether a stated persona can outweigh an explicit arithmetic cue.)
- (b) **(9 pts)** Design the diagnostic simulation: propose at least two prompt-level manipulations (e.g. removing the explicit numbers and describing returns qualitatively; adding a concrete competing use for the \$10 such as school fees due this week; burying the decision in a longer narrative) and state what result pattern under each manipulation would tell you. Run it if API budget allows and report; otherwise state the predicted patterns precisely.
- (c) **(8 pts)** The research group planned to use this sim to pick which treatment arms deserve budget in the human experiment (e.g. “timing didn’t matter in the sim, so drop the DKR arm”). Explain why that inference is unsafe, state what the sim CAN still safely inform, and give the design change this implies for the human pilot plan.

Attach your AI transcripts and, if you ran code, the results CSV.